

TASK 5 - CAPSTONE PROJECT

Project Title - Mini SIEM (Security Information & Event Management)
with ELK Stack.

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Environment: Authorized virtual cybersecurity laboratory.

Monitoring stack: Filebeat, Logstash, Elasticsearch, Kibana.

Security Tools: Kali Linux.

1.Executive Summary:

This capstone project focuses on the development of a Mini Security Information and Event Management (SIEM) system using the ELK Stack. The project demonstrates how security logs can be generated, collected, processed, indexed, searched, and visualized in a centralized monitoring environment.

The project uses Docker to deploy Elasticsearch, Logstash, Filebeat, and Kibana. Simulated security events were stored in a security log file and processed through the SIEM pipeline. Elasticsearch was used to store and verify the collected events, while Kibana was used to create a data view and visualizations for security-event analysis.

The completed system successfully indexed 12 security events. Kibana visualizations were created to display events by severity and event type, providing a centralized dashboard for monitoring security activity.

The project was performed in a controlled local laboratory environment.

2. Project Objectives:

The main objective of the project are:

1. To understand the basic architecture of a SIEM system.
2. To deploy an ELK-based monitoring environment using Docker.
3. To generate simulated security events.
4. To collect and process security logs.
5. To store security events in Elasticsearch.
6. To create a Kibana data view for the collected events.
7. To analyze security events using Kibana.
8. To create visualizations based on event severity and event type.
9. To verify the number and contents of indexed security events.
10. To demonstrate a basic incident detection and response workflow.

3. Project Scope:

The scope of the project is limited to a controlled Mini SIEM laboratory environment.

The project covers:

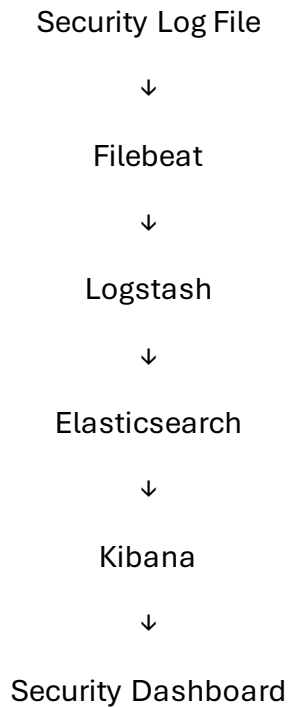
1. Docker-based deployment of ELK components.
2. Creation of simulated security logs.
3. Log collection and processing.
4. Elasticsearch indexing.
5. Kibana data visualization.
6. Security event analysis.
7. Simulated incident detection.
8. Basic containment and eradication procedures.
9. Verification of collected security events.
10. Final documentation and presentation.
11. No unauthorized external systems were tested.

4. Tools and Technologies:

Tool	Purpose
Kali Linux	Security testing and analysis
VirtualBox	Virtualized laboratory
Docker	Containerized deployment of the SIEM components
Docker Compose	Managing the ELK containers
Filebeat	Collecting and forwarding logs
Logstash	Processing security events
Elasticsearch	Indexed event storage and search
Kibana	SIEM visualization
Linux/Bash	Managing logs and executing commands
curl	Verifying Elasticsearch connectivity and indexed events.

5. Project Architecture:

The Mini SIEM follows this general data flow:



The architecture allows security events to move from the original log source through the collection and processing layers into Elasticsearch, where they can be searched and analyzed through Kibana.

6. ER Diagram – Mini SIEM (ELK Stack):

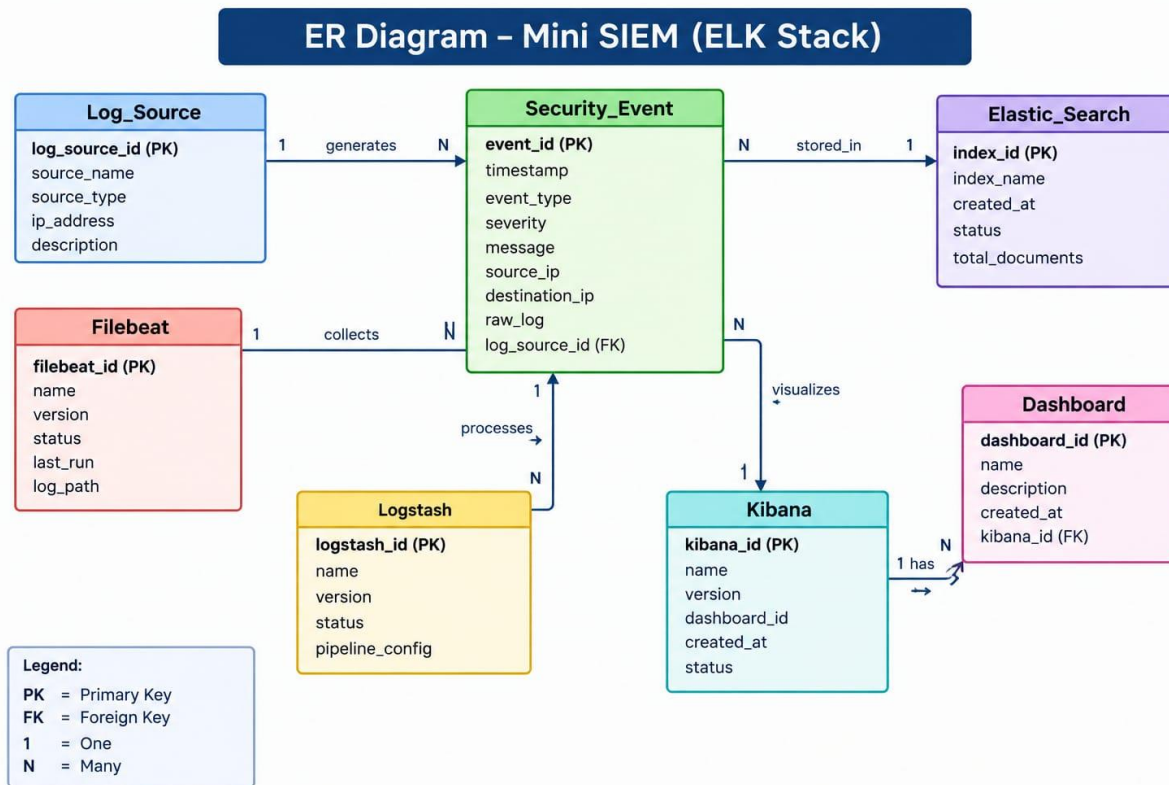


Figure 1: Network and system architecture of the Mini SIEM using the ELK Stack.

7. Methodology:

The project followed a structured SIEM implementation and incident-response methodology. The work was performed in a controlled local laboratory environment using Docker and the ELK Stack.

Phase 1: Planning and Environment Preparation

The project objectives, scope, required tools, and implementation of workflow were defined. Docker and Docker Compose were verified before deploying the SIEM components.

Evidence: Figure 1.

Phase 2: Security Log Generation

A security log file was prepared containing simulated security events. This log file serves as the source of data for the Mini SIEM.

Evidence: Figure 2.

Phase 3: SIEM Deployment

The ELK Stack components were deployed using Docker Compose. The status of Elasticsearch, Filebeat, Kibana, and Logstash containers was checked to verify the environment.

Evidence: Figure 3.

Phase 4: Elasticsearch Verification

Elasticsearch connectivity was tested using cURL. The Elasticsearch indices were then inspected to verify that the SIEM event index was available.

Evidence: Figure 4 and 5.

Phase 5: Kibana Configuration

Kibana was accessed through the browser, and the SIEM Events data view was configured. The time range was set to Last 7 days to analyze the available security events.

Evidence: Figure 6 and 7.

Phase 6: Security Event Visualization

Kibana visualizations were created to analyze the collected events. The first visualization represented events according to severity, while the second represented events according to event type. Both visualizations were added to the Mini SIEM Security Dashboard.

Evidence: Figure 8 and 9.

Phase 7: Event Verification

The total number of indexed events was verified directly through the Elasticsearch `_count` API. The search API was then used to retrieve the indexed event documents for detailed verification. The Elasticsearch count confirmed 12 security events.

Evidence: Figure 10 and 11.

Phase 8: Incident Response Simulation

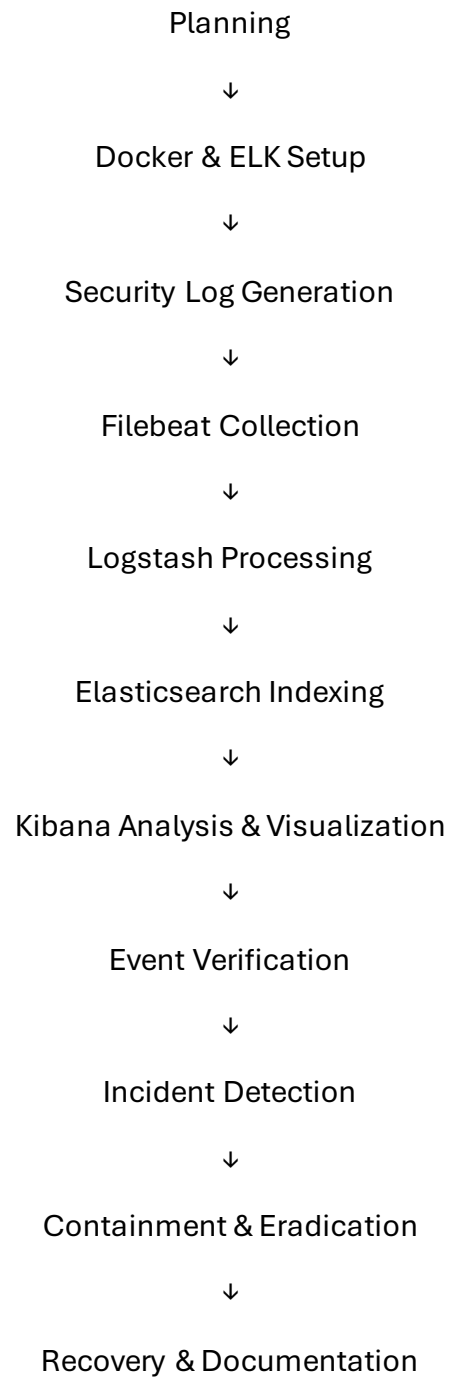
The collected simulated security events were treated as an incident-response scenario.

The response methodology consisted of:

Detection → Analysis → Containment → Eradication → Recovery

During detection and analysis, the events were reviewed through Kibana and Elasticsearch. Containment involved identifying and stopping the simulated suspicious activity. Eradication involved reviewing and correcting the relevant security configuration. Recovery involved verifying that the SIEM services and event-collection pipeline continued to operate correctly.

Methodology Flow



8. Implementation:

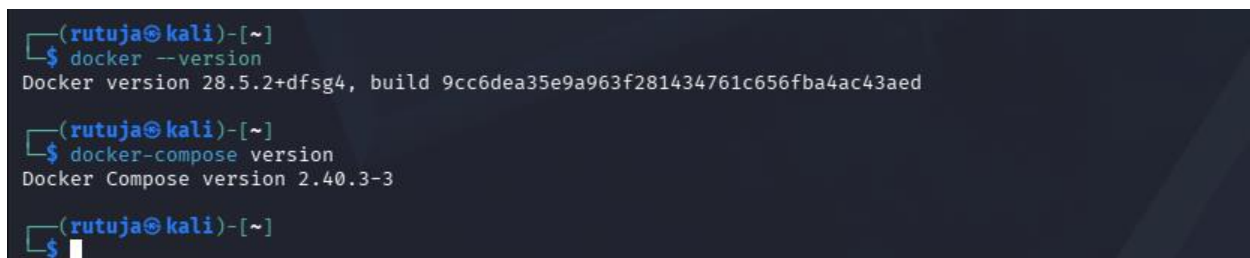
8.1 Docker Environment Verification

The first step was to verify that Docker and Docker Compose were installed and available in the Linux environment.

The following commands were used:

- `docker --version`
- `docker-compose version`

The commands returned to the installed Docker and Docker Compose versions.



```
(rutuja@kali)-[~]
$ docker --version
Docker version 28.5.2+dfsg4, build 9cc6dea35e9a963f281434761c656fba4ac43aed

(rutuja@kali)-[~]
$ docker-compose version
Docker Compose version 2.40.3-3

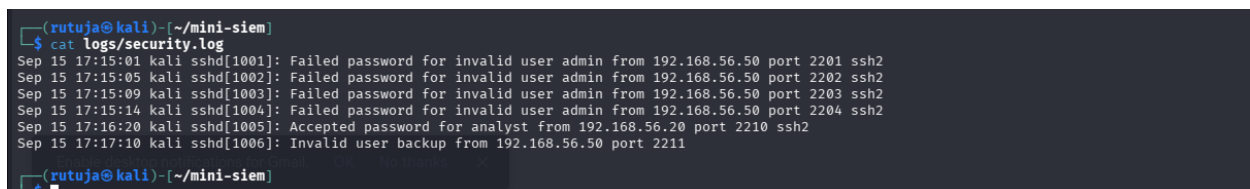
(rutuja@kali)-[~]
$
```

Figure 2: Verification of Docker and Docker Compose versions used for the Mini SIEM environment.

9. Security Log Creation:

A security log file was used as the source of simulated security events.

The log file was viewed from the terminal to verify that security-event data was available for processing by the SIEM pipeline.



```
(rutuja@kali)-[~/mini-siem]
$ cat logs/security.log
Sep 15 17:15:01 kali sshd[1001]: Failed password for invalid user admin from 192.168.56.50 port 2201 ssh2
Sep 15 17:15:05 kali sshd[1002]: Failed password for invalid user admin from 192.168.56.50 port 2202 ssh2
Sep 15 17:15:09 kali sshd[1003]: Failed password for invalid user admin from 192.168.56.50 port 2203 ssh2
Sep 15 17:15:14 kali sshd[1004]: Failed password for invalid user admin from 192.168.56.50 port 2204 ssh2
Sep 15 17:16:20 kali sshd[1005]: Accepted password for analyst from 192.168.56.20 port 2210 ssh2
Sep 15 17:17:10 kali sshd[1006]: Invalid user backup from 192.168.56.50 port 2211

(rutuja@kali)-[~/mini-siem]
$
```

Figure 3: Security log file containing simulated security events used as the Mini SIEM log source.

10. ELK Container Verification:

Docker Compose was used to manage the SIEM components.

The running containers were checked to verify the status of the Elasticsearch, Filebeat, Kibana, and Logstash services.

The main SIEM services were:

- siem-elasticsearch
- siem-filebeat
- siem-kibana
- siem-logstash

```
(rituja@kali) ~/mini-siem
$ docker-compose ps

```

NAME	IMAGE	COMMAND	SERVICE	CREATED	STATUS	PORTS
siem-elasticsearch	docker.elastic.co/elasticsearch/elasticsearch:9.5.3	"/bin/tini -- /usr/L..."	elasticsearch	2 minutes ago	Up 2 minutes	0.0.0.0:9200→9200/tcp, [::]:9200→9200/tcp, 9300/tcp
siem-filebeat	docker.elastic.co/beats/filebeat:9.5.3	"/usr/bin/tini -- /u..."	filebeat	2 minutes ago	Up 16 seconds	
siem-kibana	docker.elastic.co/kibana/kibana:9.5.3	"/bin/tini -- /usr/L..."	kibana	2 minutes ago	Up 2 minutes	0.0.0.0:5601→5601/tcp, [::]:5601→5601/tcp
siem-logstash	docker.elastic.co/logstash/logstash:9.5.3	"/usr/local/bin/dock..."	logstash	2 minutes ago	Up 2 minutes	0.0.0.0:5044→5044/tcp, [::]:5044→5044/tcp, 9600/tcp

```
(rituja@kali) ~/mini-siem
$
```

Figure 4: Docker Compose status showing the Mini SIEM ELK containers.

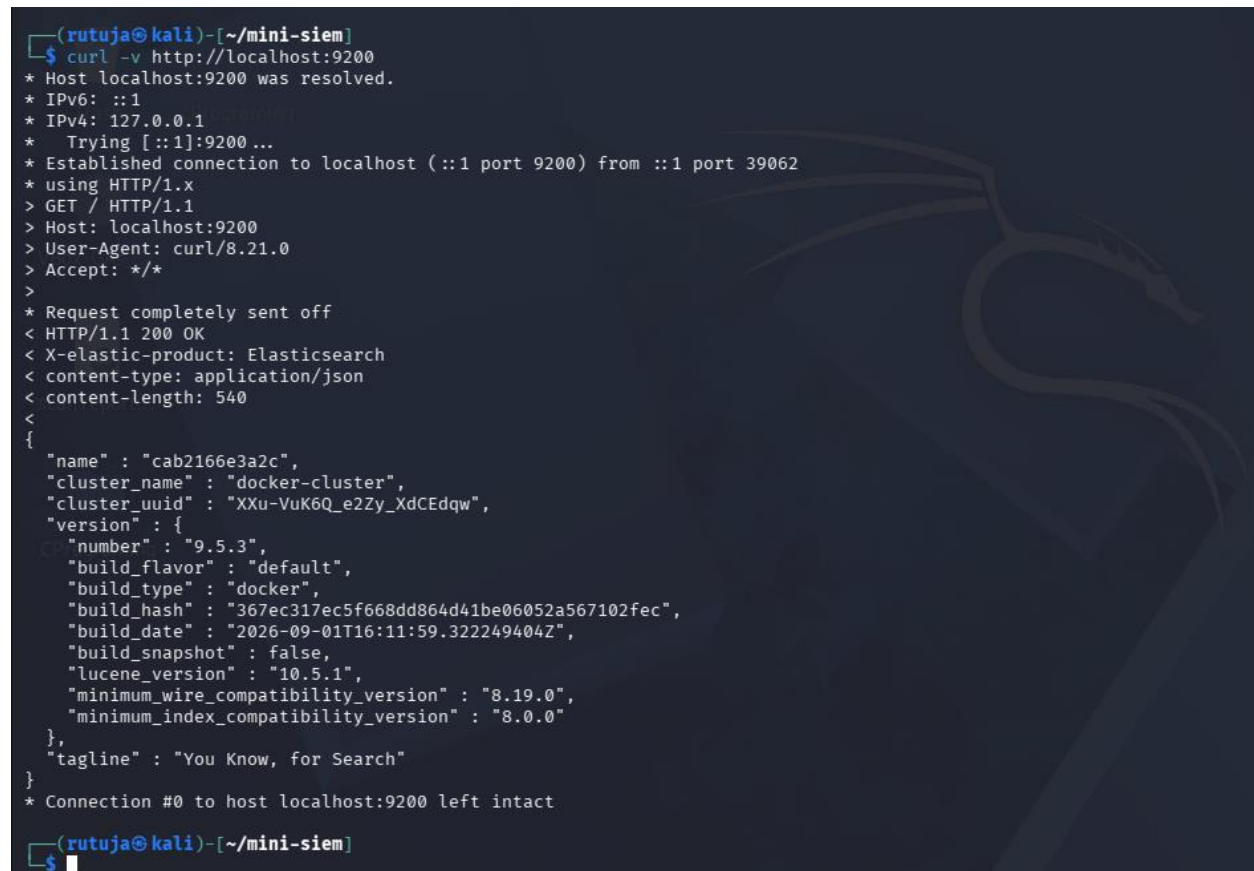
11. Elasticsearch Connectivity Test:

Elasticsearch was tested using cURL to verify communication with the Elasticsearch service.

The following command was used:

```
curl -v http://localhost:9200
```

This command was used to inspect the HTTP connection to Elasticsearch and verify service communication.



```
(rituja@kali)-[~/mini-siem]
$ curl -v http://localhost:9200
* Host localhost:9200 was resolved.
* IPv6: ::1
* IPv4: 127.0.0.1
* Trying [::1]:9200 ...
* Established connection to localhost (::1 port 9200) from ::1 port 39062
* using HTTP/1.x
> GET / HTTP/1.1
> Host: localhost:9200
> User-Agent: curl/8.21.0
> Accept: */*
>
* Request completely sent off
< HTTP/1.1 200 OK
< X-elastic-product: Elasticsearch
< content-type: application/json
< content-length: 540
<
{
  "name" : "cab2166e3a2c",
  "cluster_name" : "docker-cluster",
  "cluster_uuid" : "XXu-VuK6Q_e2Zy_XdCEdqw",
  "version" : {
    "number" : "9.5.3",
    "build_flavor" : "default",
    "build_type" : "docker",
    "build_hash" : "367ec317ec5f668dd864d41be06052a567102fec",
    "build_date" : "2026-09-01T16:11:59.322249404Z",
    "build_snapshot" : false,
    "lucene_version" : "10.5.1",
    "minimum_wire_compatibility_version" : "8.19.0",
    "minimum_index_compatibility_version" : "8.0.0"
  },
  "tagline" : "You Know, for Search"
}
* Connection #0 to host localhost:9200 left intact
(rituja@kali)-[~/mini-siem]
$
```

Figure 5: Elasticsearch connectivity verification using cURL.

12. Elasticsearch Index Verification:

The Elasticsearch `_cat/indices` API was used to inspect the available indices.

The command:

`curl -v http://localhost:9200/_cat/indices?v`

was used to verify that the SIEM event index was available in Elasticsearch.

The resulting output showed the security-event index and its document information.

```
(rituja@kali)-[~/mini-siem]
└─$ curl http://localhost:9200/_cat/indices?v
health status index      uuid                                pri rep docs.count docs.deleted store.size pri.store.size dataset.size
green  open   .internal.alerts-security.alerts-default-000001 E8j7yBmDTxGcyz24wiUYww  1  0         0             0        249b          249b          249b
yellow open    siem-events-2026.09.16  8N_n1BvkSPmZ5M1eT7SaBg  1  1         12             0       74.1kb          74.1kb          74.1kb

(rituja@kali)-[~/mini-siem]
└─$
```

Figure 6: Elasticsearch index verification showing the SIEM security-event index.

13. Kibana Environment:

After verifying the backend services, Kibana was opened through the web browser.

The Elastic welcome page provided access to the main Elastic components, including:

- Elasticsearch
- Observability
- Security
- Analytics

This confirmed that the Kibana interface was accessible and ready for SIEM analysis.

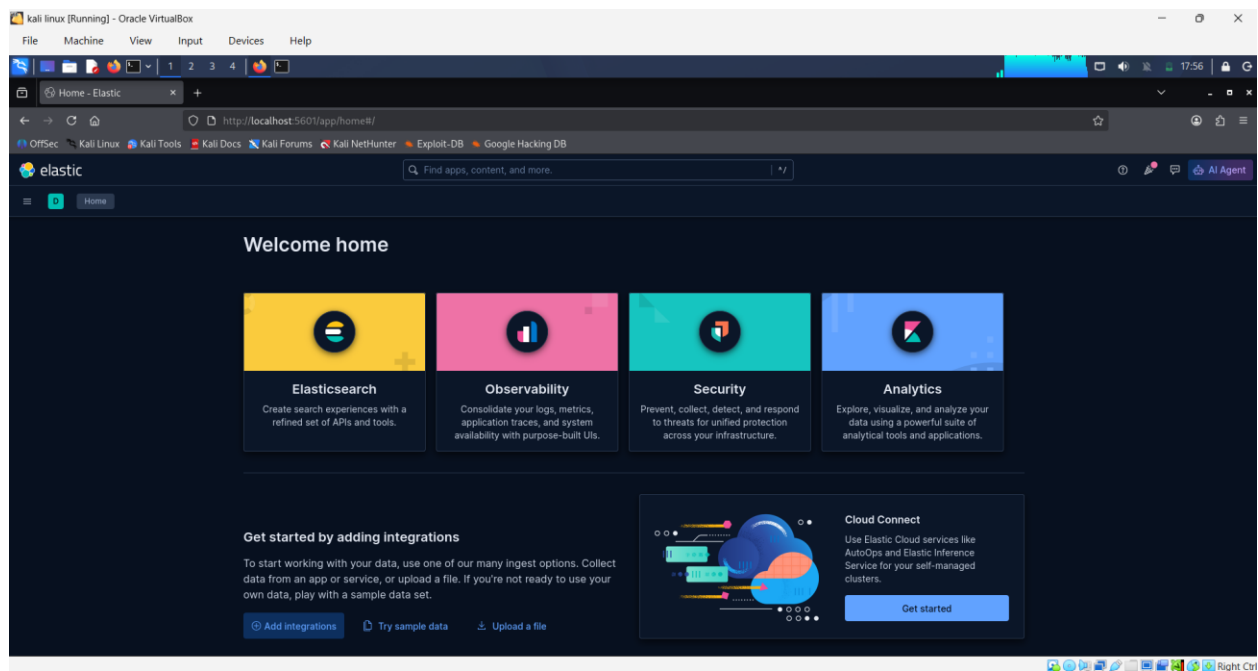


Figure 7: Elastic welcome page displaying the available Elasticsearch, Observability, Security, and Analytics features.

14. Kibana Data View and Security Events:

A Kibana data view named SIEM Events was configured for the security-event index.

The time range was set to Last 7 days to display the relevant security events.

The Discover interface provided access to:

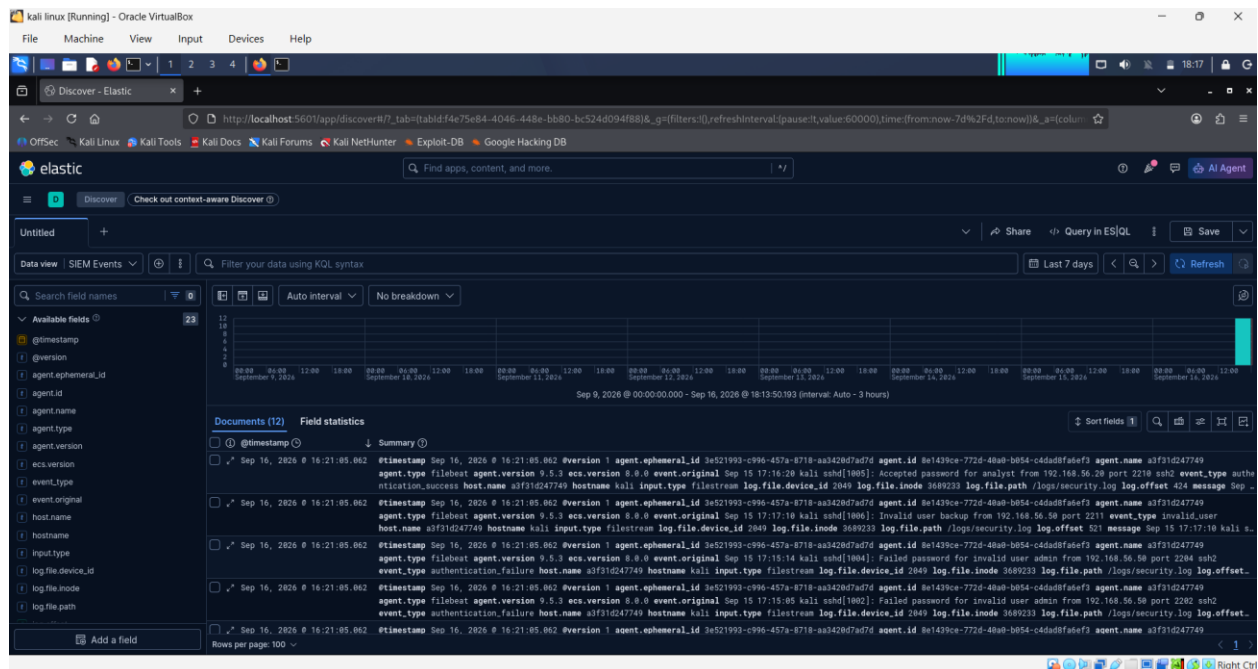


Figure 8: Kibana Discover view

15. Events by Severity Visualization:

A Kibana visualization was created to analyze security events according to their severity.

The visualization uses the severity.keyword field to categorize the events. The resulting chart displays categories such as:

- High
- Medium
- Low

This visualization provides a quick overview of the severity distribution of the collected security events.

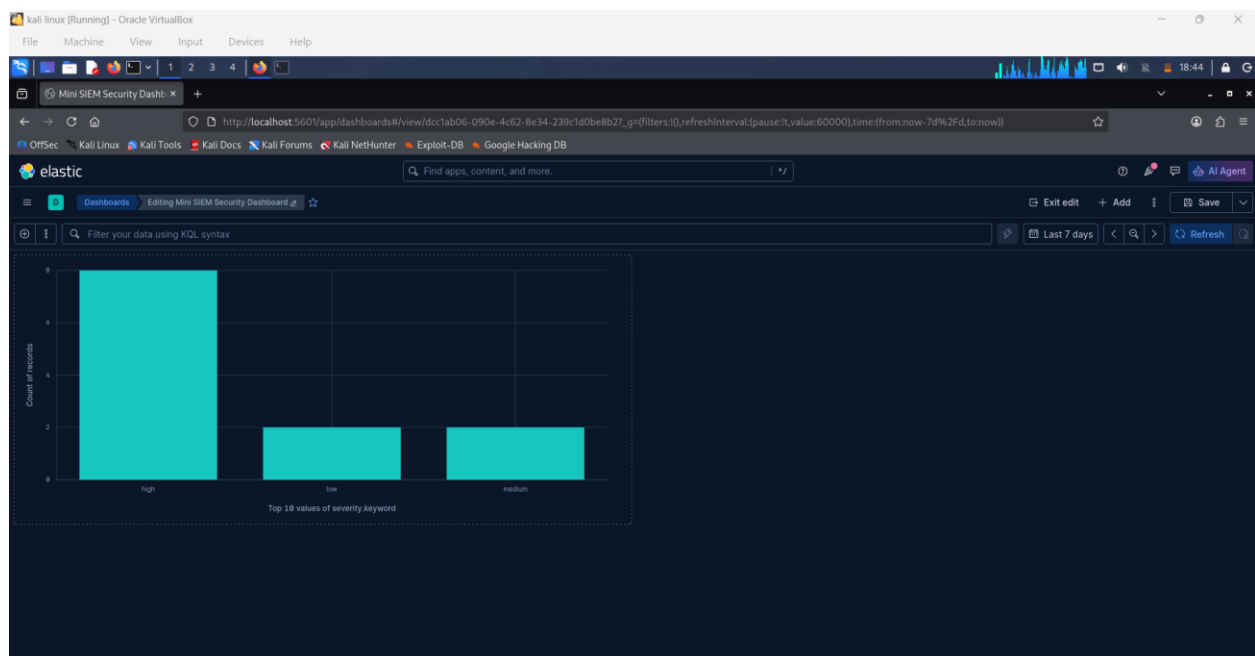


Figure 9: Kibana visualization showing security events categorized by severity.

16. Events by Type Visualization:

A second visualization was created to analyze the security events according to their event type.

The events-by-type visualization provides another perspective of the collected data and helps identify the different types of security activity recorded by the Mini SIEM.

Both visualizations were added to the Mini SIEM Security Dashboard.

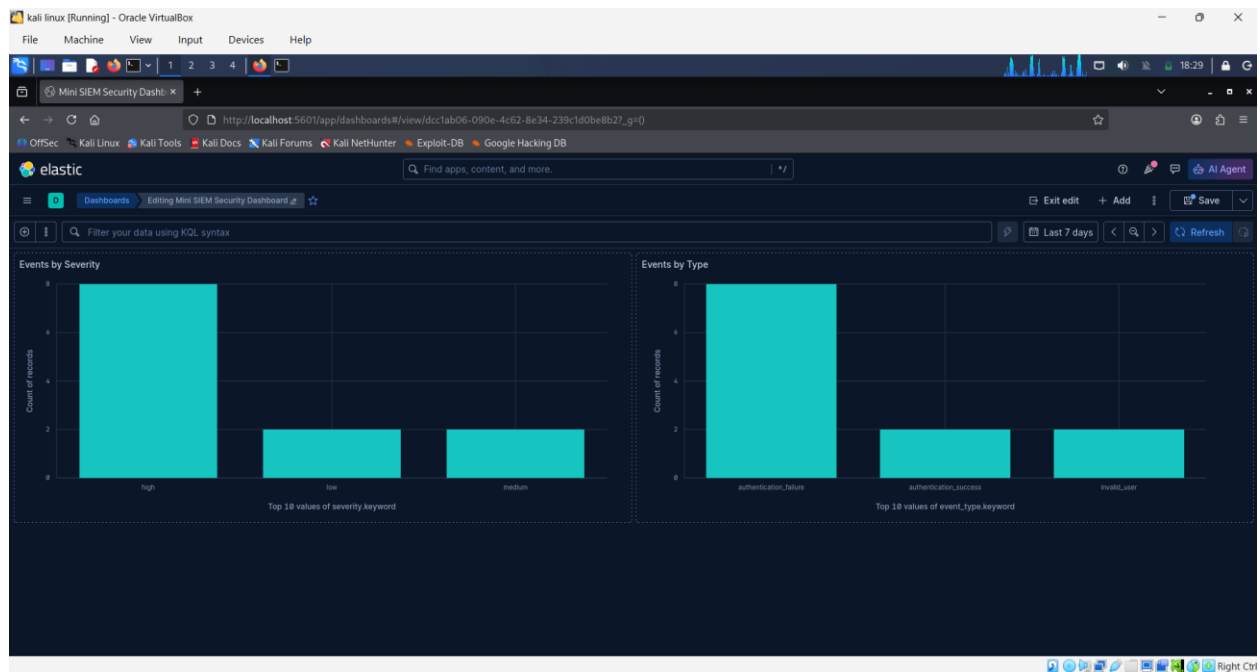


Figure 10: Mini SIEM Security Dashboard showing events by severity and events by type.

17. Elasticsearch Event Count Verification:

After completing the Kibana dashboard, Elasticsearch was queried to verify the total number of security events stored in the SIEM index.

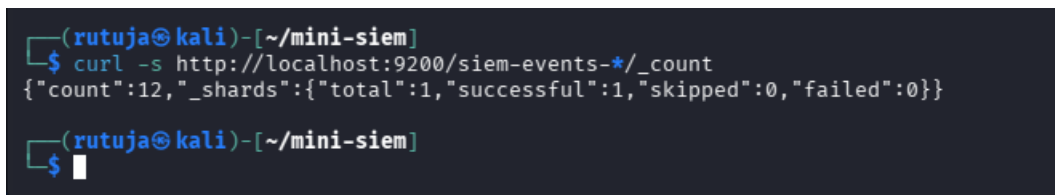
The following command was executed:

```
curl -s http://localhost:9200/siem-events-\*/\_count
```

The result confirmed:

Count: 12

This verified that 12 security events were indexed in Elasticsearch.

A terminal window with a dark background. The prompt is (rutuja@kali) - [~/mini-siem]. The user enters the command curl -s http://localhost:9200/siem-events-*/_count. The output is {"count":12,"_shards":{"total":1,"successful":1,"skipped":0,"failed":0}}. The user enters another prompt (rutuja@kali) - [~/mini-siem] and a dollar sign followed by a space.

```
(rutuja@kali) - [~/mini-siem]
$ curl -s http://localhost:9200/siem-events-*/_count
{"count":12,"_shards":{"total":1,"successful":1,"skipped":0,"failed":0}}
(rutuja@kali) - [~/mini-siem]
$
```

Figure 11: Elasticsearch count API confirming 12 indexed SIEM security events.

18. Elasticsearch Event Data Verification:

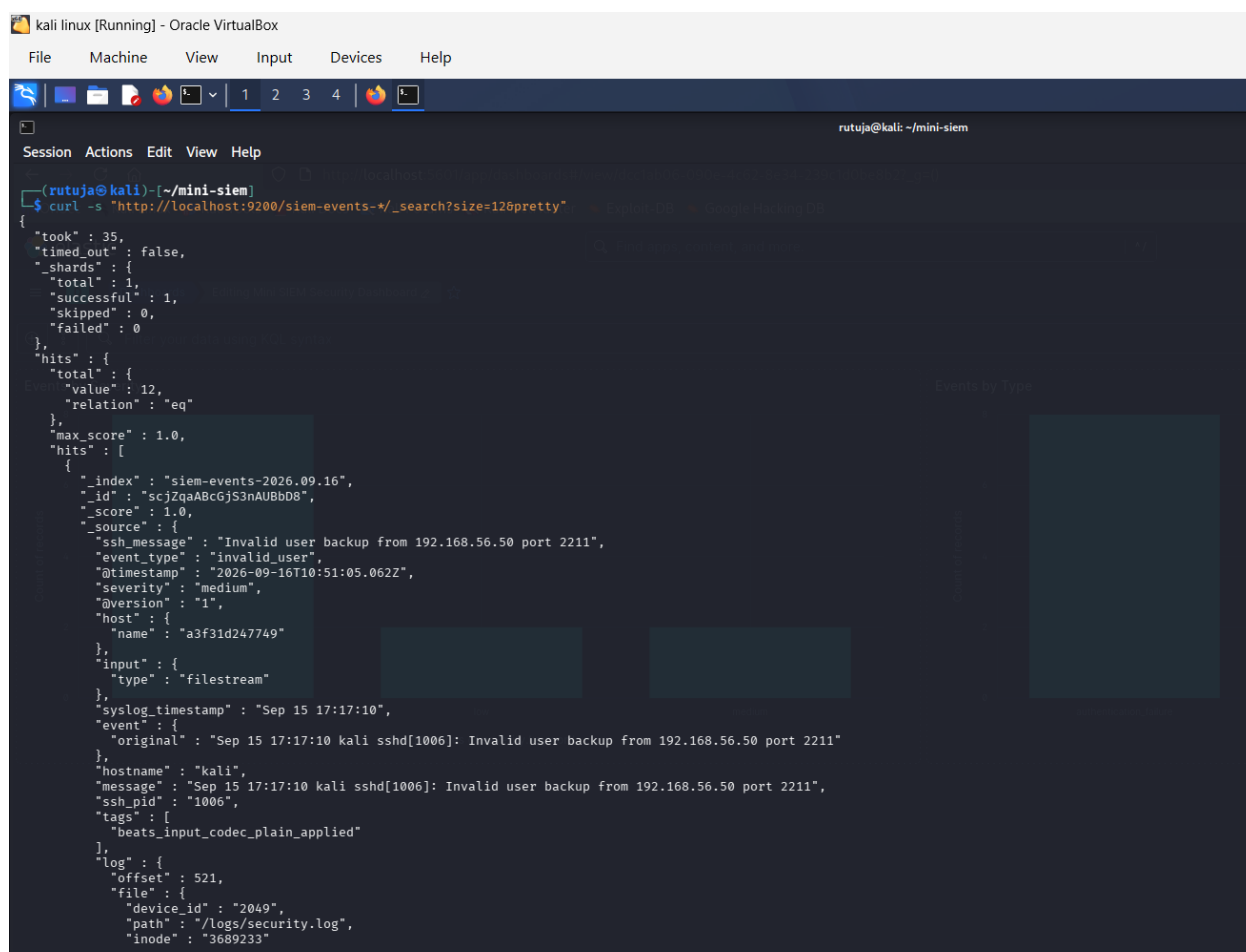
The indexed security events were then retrieved directly from Elasticsearch using the `_search` API.

The following command was used:

```
curl -s http://localhost:9200/siem-events-*/_search?size=12&pretty
```

The command returned the stored security-event documents in formatted JSON output.

This provided direct evidence that the security events were successfully stored in Elasticsearch and could be retrieved for investigation.



```
kali linux [Running] - Oracle VirtualBox
File Machine View Input Devices Help

(rutuja@kali) - [~/mini-siem]
$ curl -s "http://localhost:9200/siem-events-*/_search?size=12&pretty"
{
  "took" : 35,
  "timed_out" : false,
  "_shards" : {
    "total" : 1,
    "successful" : 1,
    "skipped" : 0,
    "failed" : 0
  },
  "hits" : {
    "total" : {
      "value" : 12,
      "relation" : "eq"
    },
    "max_score" : 1.0,
    "hits" : [
      {
        "_index" : "siem-events-2026.09.16",
        "_id" : "scj2qaABcGjS3nAU8bD08",
        "_score" : 1.0,
        "_source" : {
          "ssh_message" : "Invalid user backup from 192.168.56.50 port 2211",
          "event_type" : "invalid_user",
          "@timestamp" : "2026-09-16T10:51:05.062Z",
          "severity" : "medium",
          "@version" : "1",
          "host" : {
            "name" : "a3f31d247749"
          },
          "input" : {
            "type" : "filestream"
          },
          "syslog_timestamp" : "Sep 15 17:17:10",
          "event" : {
            "original" : "Sep 15 17:17:10 kali sshd[1006]: Invalid user backup from 192.168.56.50 port 2211"
          },
          "hostname" : "kali",
          "message" : "Sep 15 17:17:10 kali sshd[1006]: Invalid user backup from 192.168.56.50 port 2211",
          "ssh_pid" : "1006",
          "tags" : [
            "beats_input_codec_plain_applied"
          ],
          "log" : {
            "offset" : 521,
            "file" : {
              "device_id" : "2049",
              "path" : "/logs/security.log",
              "inode" : "3689233"
            }
          }
        }
      }
    ]
  }
}
```

Figure 12: Elasticsearch search API displaying the indexed security-event documents.

19. Incident Response Simulation:

19.1 Incident Scenario

For the incident-response portion of the project, simulated security events were generated and collected by the Mini SIEM.

The purpose of the simulation was to demonstrate how a security analyst could identify suspicious activity from centralized logs.

The events were analyzed using the SIEM pipeline:

Log Source → Filebeat → Logstash → Elasticsearch → Kibana

20. Incident Detection:

The simulated security activity was detected by analyzing the collected security events.

Kibana provided a centralized interface for reviewing the events, while Elasticsearch provided direct access to the indexed event data.

The completed dashboard displayed security events according to severity and event type.

The Elasticsearch count query further confirmed that 12 events were available for analysis.

21. Incident Containment:

The following containment actions were considered for the simulated incident:

1. Identify the suspicious security event.
2. Review the event details and timestamps.
3. Identify the affected log source.
4. Isolate the affected component if required.
5. Stop or restrict the simulated suspicious activity.
6. Preserve the relevant security logs for investigation.

22. Incident Eradication:

The following eradication steps were considered:

1. Identify the cause of the simulated suspicious activity.
2. Remove the simulated unwanted activity.
3. Review the affected configuration.
4. Apply appropriate security controls.
5. Verify that the suspicious activity no longer continues.
6. Continue monitoring the environment through the SIEM.

23. Recovery:

The recovery process included:

1. Verifying that the ELK services were operational.
2. Checking Elasticsearch connectivity.
3. Confirming that security logs were still being collected.
4. Verifying that events continued to appear in Elasticsearch.
5. Checking the Kibana dashboard.
6. Confirming the final event count.

The completed environment successfully displayed and retrieved the collected security events.

24. Findings:

ID	Finding	Evidence
F-01	Security events were successfully generated and stored in a log file.	Screenshot 2
F-02	ELK containers were successfully deployed and running.	Screenshot 3
F-03	Elasticsearch connectivity was verified.	Screenshot 4
F-04	SIEM event indexing was confirmed in Elasticsearch.	Screenshot 5
F-05	Kibana was successfully accessed and configured for analysis.	Screenshot 6-7
F-06	Security events could be visualized by severity.	Screenshot 8
F-07	Security events could be visualized by event type.	Screenshot 9
F-08	Elasticsearch confirmed 12 index security events	Screenshot 10
F-09	Indexed event documents can be retrieved from Elasticsearch.	Screenshot 11

25. Mitigation Strategies:

Based on the SIEM monitoring workflow, the following defensive measures are recommended:

Log Monitoring

- Centralize security logs.
- Monitor authentication and security-related events.
- Maintain appropriate log-retention policies.
- Regularly review unusual event patterns.

SIEM Monitoring

- Configure alerts for suspicious activities.
- Monitor high-severity events.
- Use event-type categorization for investigation.
- Regularly review SIEM dashboards.

System Security

- Keep operating systems and services updated.
- Disable unnecessary services.
- Apply least-privilege access.
- Use strong authentication controls.

Incident Response

- Maintain an incident-response procedure.
- Preserve logs as investigation evidence.
- Document incidents and response actions.
- Verify system recovery after an incident.

26. Project Results:

The project successfully achieved the following results:

- Docker environment successfully configured.
- ELK components successfully deployed.
- Security log source successfully created.
- Elasticsearch connectivity verified.
- SIEM index successfully identified.
- Kibana successfully accessed.
- SIEM Events data view successfully configured.
- Time range configured to Last 7 days.
- Events-by-severity visualization successfully created.
- Events-by-type visualization successfully created.
- Mini SIEM Security Dashboard successfully created.
- Elasticsearch confirmed 12 security events.
- Indexed event documents were successfully retrieved.

27. Conclusion:

The Mini SIEM project successfully demonstrated centralized security-event monitoring using the ELK Stack. Security logs were collected, processed, indexed in Elasticsearch, and analyzed through Kibana. The project successfully created visualizations for events by severity and event type, while Elasticsearch confirmed 12 indexed security events. The project also demonstrated a basic incident-response workflow for detecting, analyzing, containing, and recovering from simulated security events.